E-news Express

Business Statistics Project

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Executive Summary

Conclusion

1. **Do users spend more time on the new landing page than on the old landing page?**

   Yes — Users spent significantly more time on the new landing page.
   (**One-Tailed Welch's Two-Sample T-test** , $p < 0.05$)

2. **Is the conversion rate for the new page greater than the conversion rate for the old page?**

   Yes — The new landing page achieved a higher conversion rate.
   (**Two Proportion Z-test**, $p < 0.05$)

3. **Does the converted status depend on the preferred language?**

   No — There is no significant relationship between language preference and conversion.
   (**Chi-Square Test of Independence**, $p > 0.05$)

4. **Is the time spent on the new page the same for different language users?**

   Yes — The average time spent is similar across English, French, and Spanish users.
   (**One-Way ANOVA test**., $p > 0.05$)

Executive Summary

Actionable Insights

- **Improved page design and content relevance** effectively boosted **user engagement** and **conversion rates**.
- **Longer time spent** on the page was associated with a **higher likelihood of conversion**.
- **User language preference** does **not meaningfully affect** engagement or conversion in the new landing page design

Executive Summary

Recommendations

- **Implement the new landing page** across the site to enhance user engagement and subscription rates.
- **Monitor and A/B test** other aspects such as **recommended content**, **visual layout**, and **call-to-action placements** (e.g., subscribe now, start free trial, etc.) to further enhance engagement.
- **Gather and analyze user demographics and profiles** (e.g., age group, gender, interests) to enable more effective segmentation and personalized targeting strategies.
- **Leverage user behavior data** to tailor content recommendations based on individual preferences and browsing patterns.
- **Explore different pricing strategies** in future experiments to assess their impact on conversion and long-term retention.
- **Track subscription duration** to evaluate how landing page experience affects user loyalty and churn over time.

Business Problem Overview and Solution Approach

Problem Statement

E-news Express, an online news portal, seeks to grow its business through acquisition of new subscribers. However, the business has seen a decline in new monthly subscribers and suspects that the current landing page design may not be engaging users effectively. A redesigned landing page was developed to improve user engagement and relevance of content. The company conducted an A/B test to evaluate whether the new landing page outperforms the old one in terms of user interaction and subscription conversion.

Business Problem Overview and Solution Approach

Objective

To evaluate the effectiveness of a newly designed landing page in boosting user engagement and conversion rates of users to subscribers.

This involves performing statistical analysis on experimental data collected from a randomized trial of 100 users (50 for the new landing page, 50 for the old landing page) to answer the following key questions:

1. Do the users **spend more time** on the **new landing page** than on the **old landing page**?
2. Is the **conversion rate** for the **new** page **greater** than the conversion rate for the **old** page?
3. Does the **converted** status **depend** on the **preferred language?**
4. Is the **time spent** on the new page the **same** for the **different language users**?

Business Problem Overview and Solution Approach

Methodology

Note: The **p-value approach** was chosen over the **critical value** as it provides more information, such as the **extent** to which there is **evidence** against the null hypothesis rather than simply indicating whether to reject it.

1. Data Collection & Cleaning
2. Exploratory Data Analysis (EDA)
   - Structure
   - Univariate
   - Bivariate
3. Hypothesis Testing
   - Defining the null and alternative hypotheses
   - Selecting the appropriate test
   - Deciding the level of significance
   - Collecting & preparing the data
   - Calculating the p-value
   - Comparing p-value with significance level
   - Drawing of inferences
4. Conclusion & Recommendations

EDA: Structure of the Data

- The data includes different attributes related to each **user** (illustrated in Table 1).
- There are a total of **100 rows (users)** and **6 columns (variables)** in the data frame.
- In terms of data types, the dataset includes **4 object** columns, **1 integer** column, and **1 float** column. Although **user_id** should be an object rather than an integer, it will not be used in tests; hence it does not require modification.
- There are **5 categorical variables** and **1 numerical (continuous) variable** in the dataset. Therefore, statistical tests involving categorical variables, as well as tests between categorical and continuous variables, are expected to be conducted.

Table 1. Data type and variable type of each user attribute.

Column Name	Data Type	Variable Type
user_id	int64	Categorical
group	object	Categorical
landing_page	object	Categorical
time_spent_on_the_page	float64	Continuous
converted	object	Categorical
language_preferred	object	Categorical

EDA: Structure of the Data

- There are **neither missing values nor duplicated values** in the data frame, so not much data cleaning and processing was required.

- **Table 2. Statistical summary of time spent on the page (numerical variable).**

Variable	count	mean	std	min	25%	50%	75%
time_spent_on_the_page	100	5.38	2.38	0.19	3.88	5.42	7.02

- **Table 3. Statistical summary of categorical variables.**

Variables	count	unique	top	freq
group	100	2	control	50
landing_page	100	2	old	50
converted	100	2	yes	54
language_preferred	100	3	Spanish	34

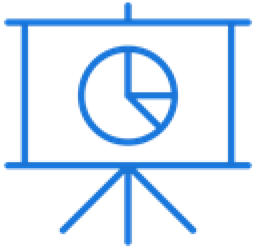
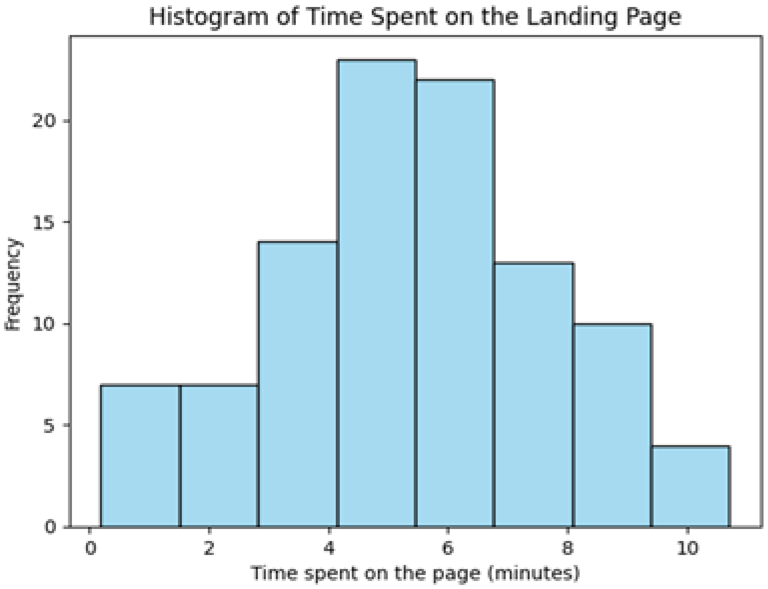
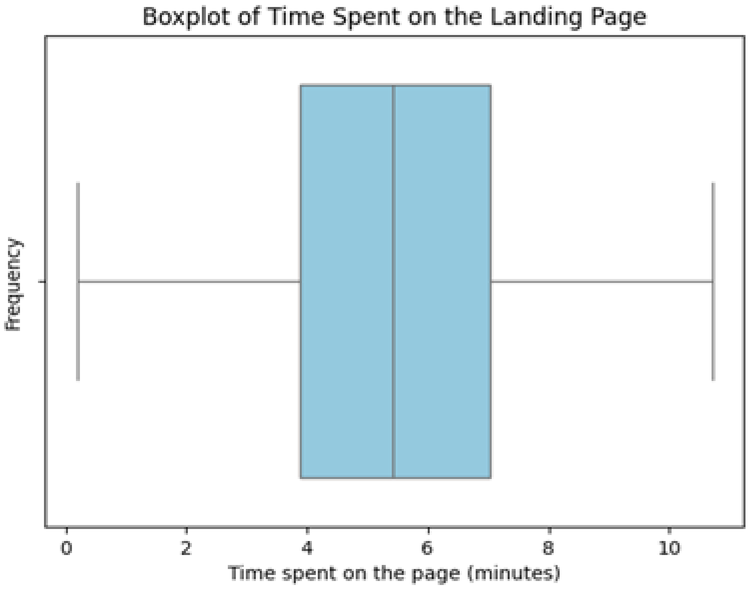


Table 3 demonstrates that the control (old landing page) and treatment (new landing page) groups each has 50 entries. In addition, 54 out of 100 users were converted to subscribers while Spanish seems to have the highest count. More will be revealed by the univariate and bivariate analyses.

EDA: Univariate Analysis - Time spent on the page

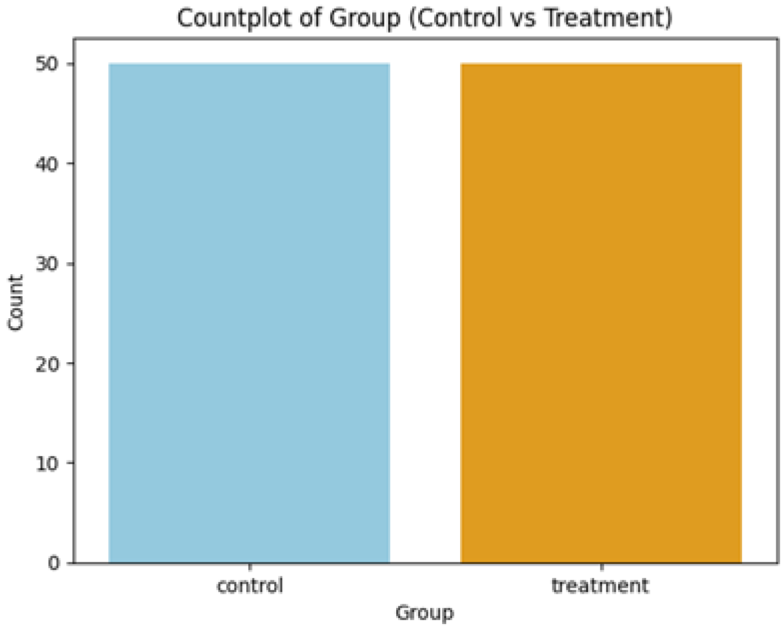


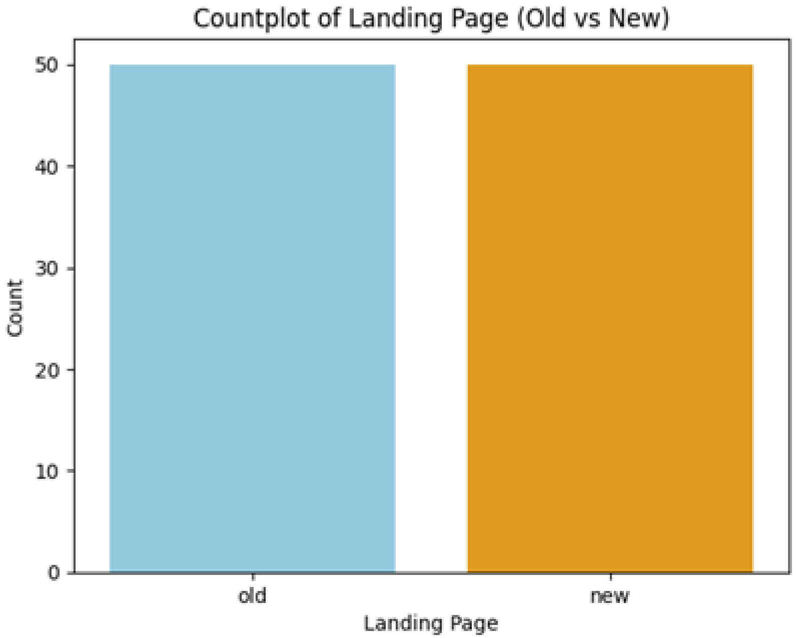


- The distribution of time spent is **roughly unimodal** and **approximately symmetric**, centered around **5–6 minutes**.
- There is **slight left-skewness** (a small tail toward lower times), but overall the distribution looks fairly close to **normal.**
- Most users spend between **4 and 7 minutes** on the page.

- The boxplot confirms the **central tendency**: the median is around **5–6 minutes**.
- There are **no extreme outliers**, suggesting consistent behavior among users.
- The interquartile range (IQR) is fairly tight, indicating **moderate variability**.

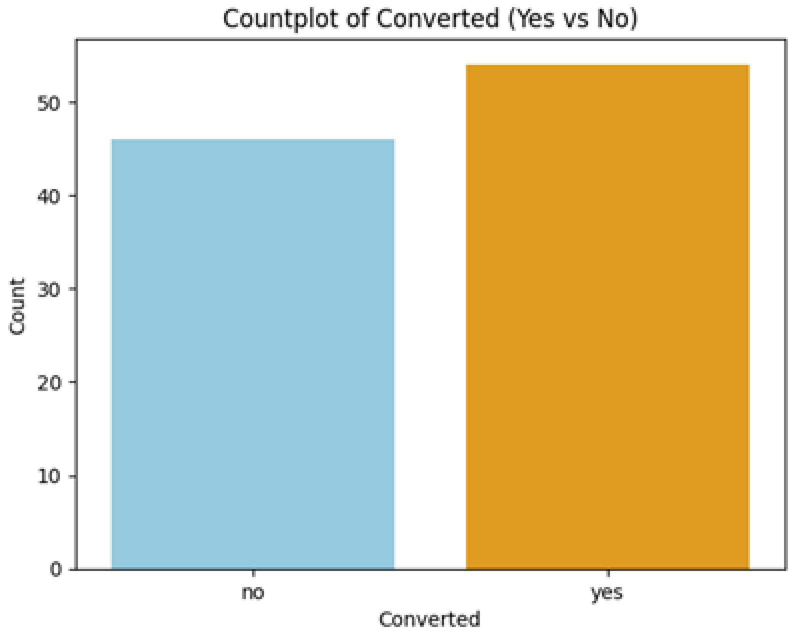
EDA: Univariate Analysis - Group & Landing Page

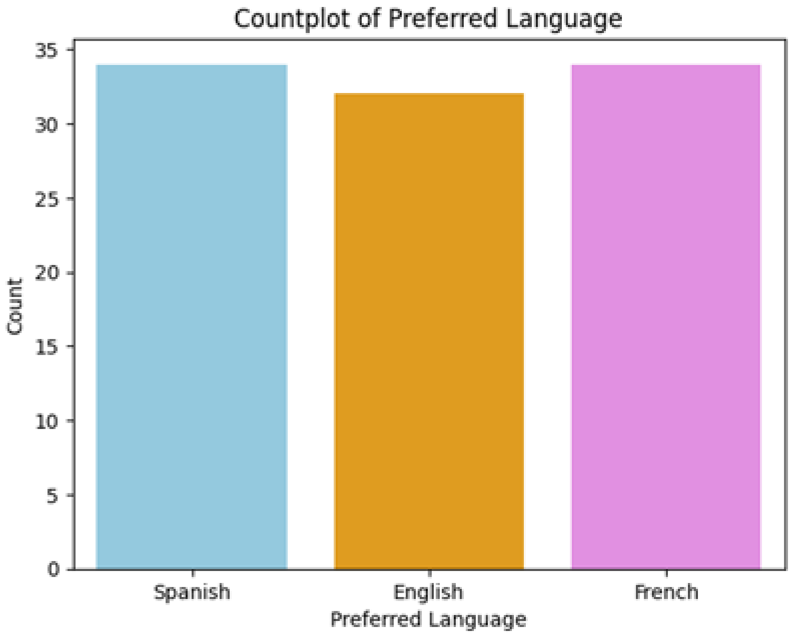




- Both count plots **confirm** that the **control (old landing page)** and **treatment (new landing page)** groups each have exactly **50 users**.
- This shows how the experiment was **well-randomized** and **evenly distributed** between the two landing page designs.

EDA: Univariate Analysis - Converted & Preferred Language

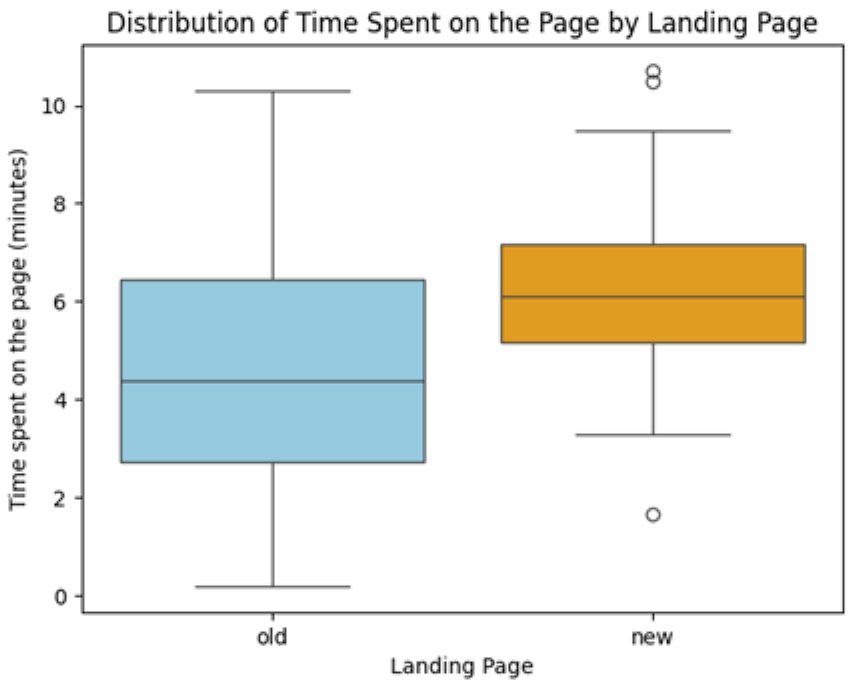




- There were **slightly more converted users into subscribers** (Yes = 54) than those who were not converted (No = 46).

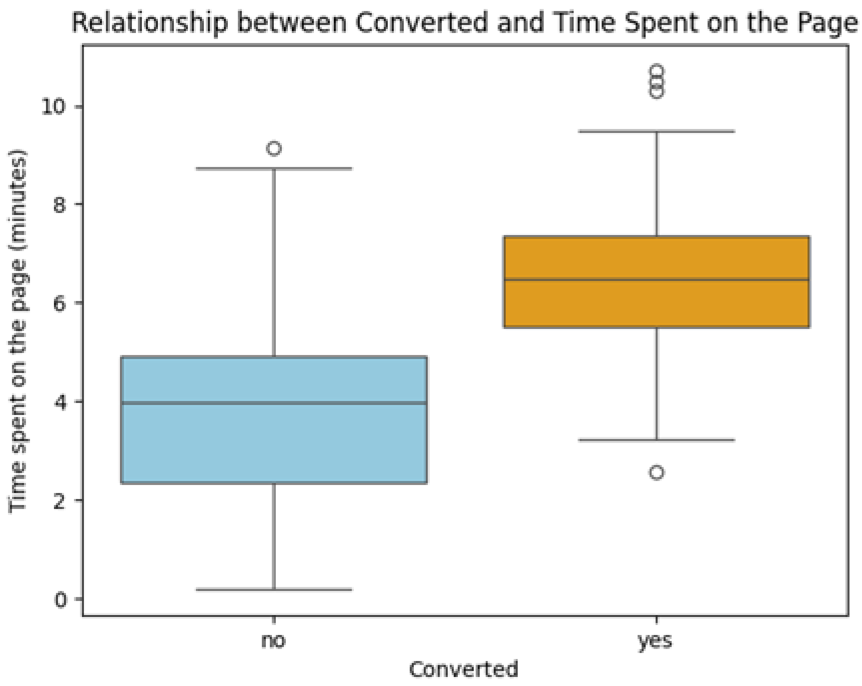
- The user counts across Spanish (34), English (32), and French (34) are quite **balanced**, allowing fair comparison across language groups.

EDA: Bivariate Analysis
Landing page vs Time spent on the page



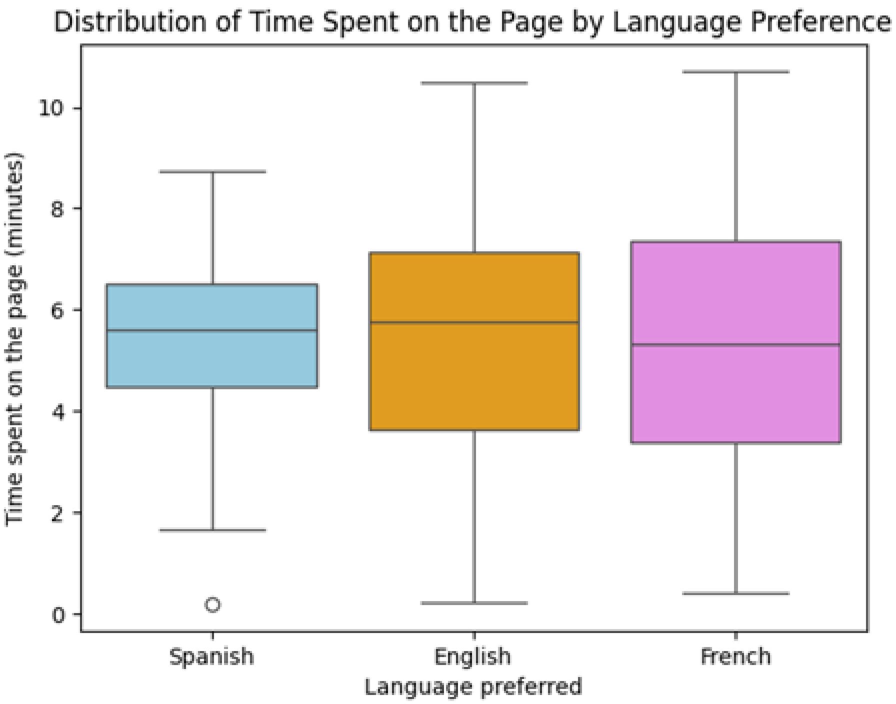
- The **median** time for the new page is **higher** than for the old page.
- The **IQR** of the new page is **smaller**, indicating **less variability.**
- The old page shows a **wider spread** of user time with some spending very little and others much more.
- A few outliers were observed in the new page, indicating how some users had **very low or very high** engagement.
- Overall, the boxplot **visually supports** that the new landing page leads to **longer and more stable user engagement**

EDA: Bivariate Analysis
Conversion status vs Time spent on the page



- The **median time** for converted users is noticeably **higher** (around 6–7 minutes) compared to non-converted users (around 4 minutes).
- The top 25% of converted users are spending **more time** than 75% of non-converted users.
- The **IQR** of the **converted users** is slightly **smaller**, indicating **less variability.**
- The time spent by **non-converted** users is **highly variable**, ranging from almost no time (close to 0) to relatively long sessions (as indicated by the extended right whisker and outlier), yet without resulting in conversion.
- Several **upper outliers** were observed in the converted users, indicating that some users spent **more than 10 minutes** with E-new's page and decided to subscribe.
- Overall, the boxplot **visually supports** that the converted users into subscribers spent more time engaging on E-new's page than those who did not.

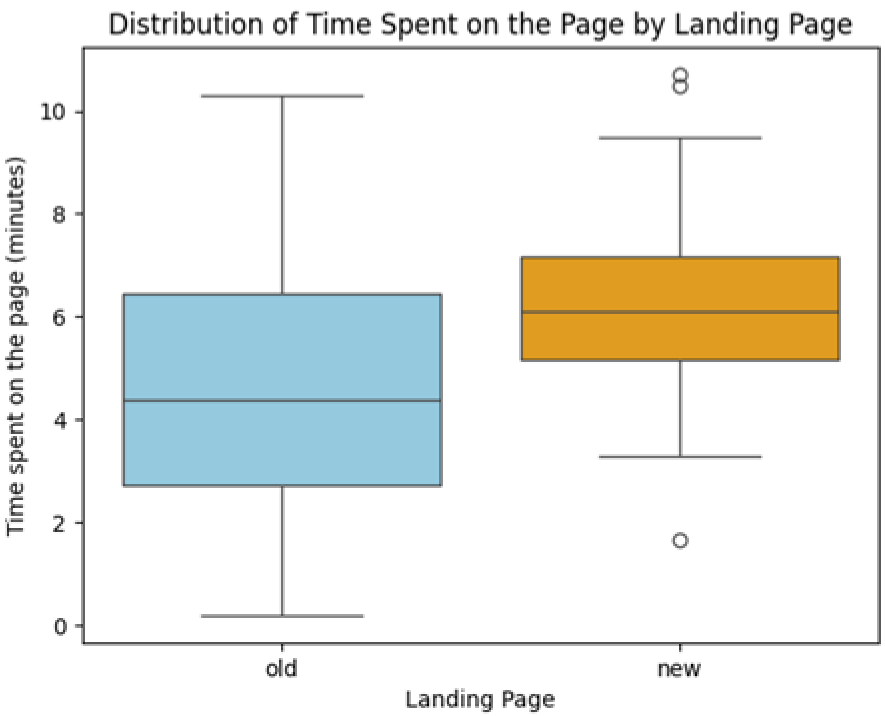
EDA: Bivariate Analysis
Language preferred vs Time spent on the page (old and new)



- The **smallest IQR** is observed for **Spanish** users, indicating more consistent time spent.
- The **largest IQR** (and thus, the most variability) is observed among **French** users.
- **English** users show the **highest median** time spent on the page, followed closely by Spanish, then French. However, the differences in medians are relatively small.
- Overall, there seems to be **little to no link between user engagement time and preferred language.** There are slight differences, but **variation within each group** (wide IQRs, overlapping ranges) is large compared to the small difference in medians.

1. Hypotheses Tested and Results

1. Do the users spend more time on the new landing page than on the old landing page?



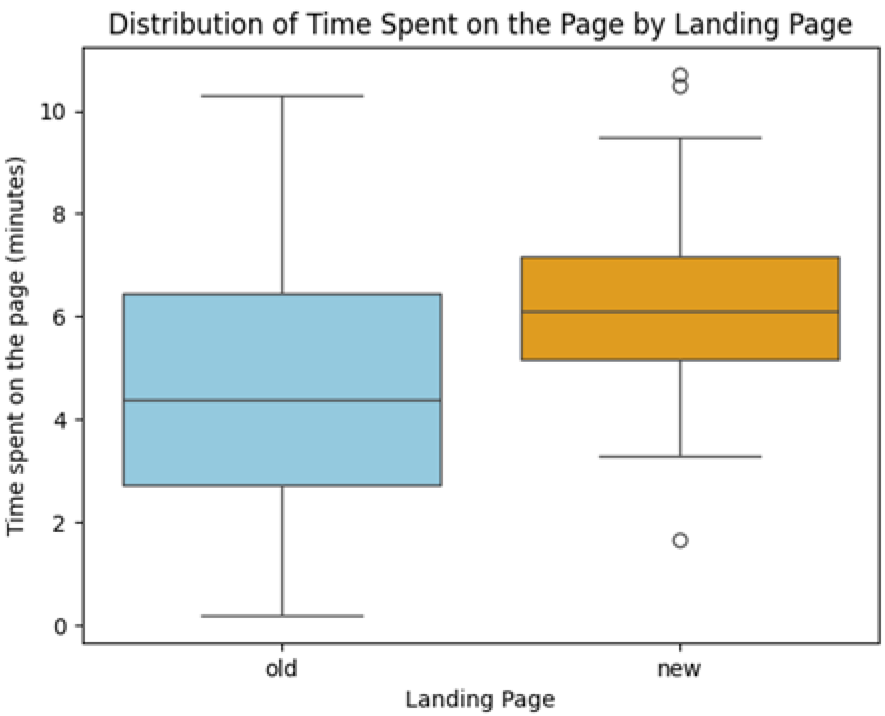
Visual Analysis

- The **median** time for the new page is **higher** than for the old page.
- The **IQR** of the new page is **smaller**, indicating **less variability.**
- The old page shows a **wider spread** of user time with some spending very little and others much more.
- A few outliers were observed in the new page, indicating how some users had **very low or very high** engagement.
- Overall, the boxplot **visually supports** that the new landing page leads to **longer and more stable user engagement**

Link to Appendix slide on details of the test performed (Slide 33)

1. Hypotheses Tested and Results

1. Do the users **spend more time** on the **new landing page** than on the **old landing page**?



Hypothesis tested

- It was tested whether users **spend more time** on the new landing page than on the old one.

Test result

- **One-tailed Welch's two-sample t-test** was conducted at a **5% significance level** ($\alpha = 0.05$).
- **p-value** = 0.000139

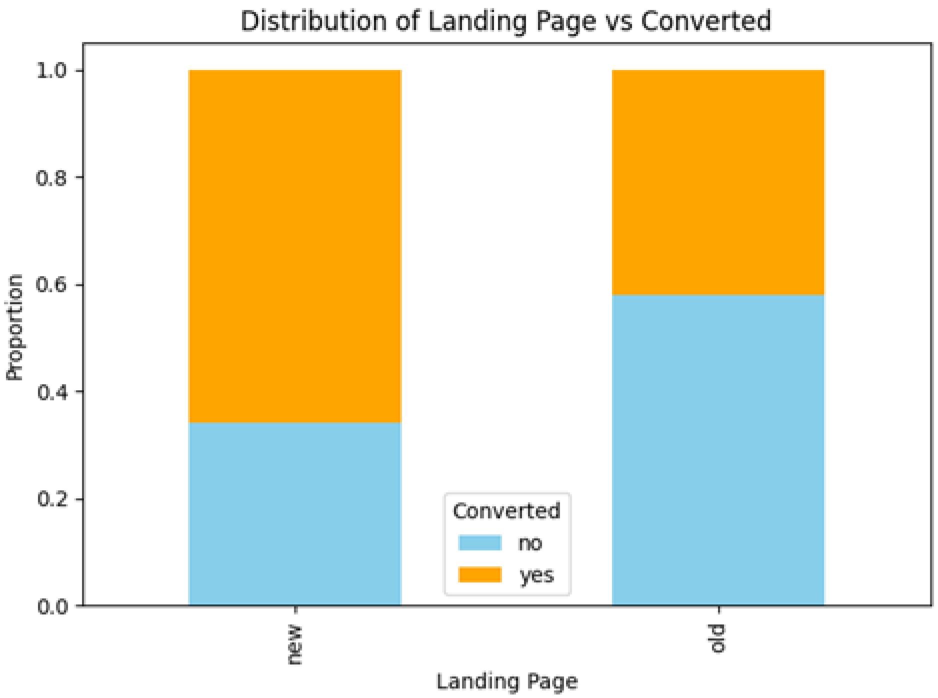
Inference

- Since the p-value (0.000139) < 0.05, there is enough evidence to **reject** the null hypothesis.
- Therefore, there is statistically significant evidence that the **new landing page leads to longer user engagement.**

Link to Appendix slide on details of the test performed (Slide 33)

2. Hypotheses Tested and Results

2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?



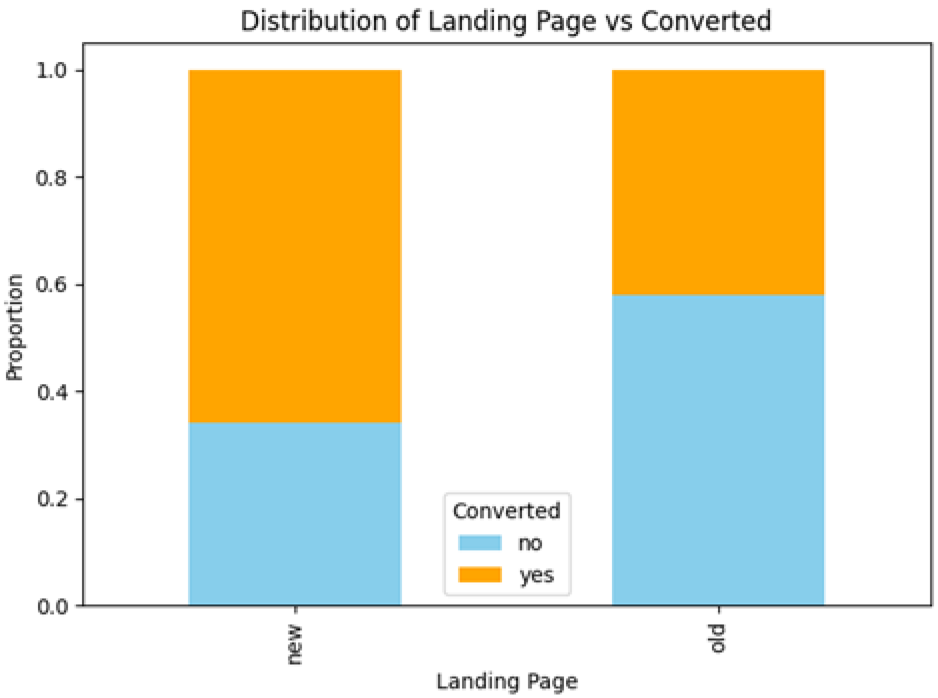
Visual Analysis

- From the stacked bar chart, the orange segment ("yes") is taller than the blue segment ("no") for the **new landing page**, suggesting a **higher conversion rate** compared to the old page.
- Conversely, the blue segment is taller than the orange segment for the **old landing page**, suggesting a **lower conversion rate.**
- This suggests that the **new landing page may be more persuasive** or better optimized for **encouraging subscriptions.**

Link to Appendix slide on details of the test performed (Slide 35)

2. Hypotheses Tested and Results

2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?



Hypothesis tested

- It was tested whether the **conversion rate** on the new landing page is **greater** than on the old landing page.

Test result

- **Two Proportion Z-test** was conducted at a **5% significance level** (α = 0.05).
- **p-value** = 0. 0.00803

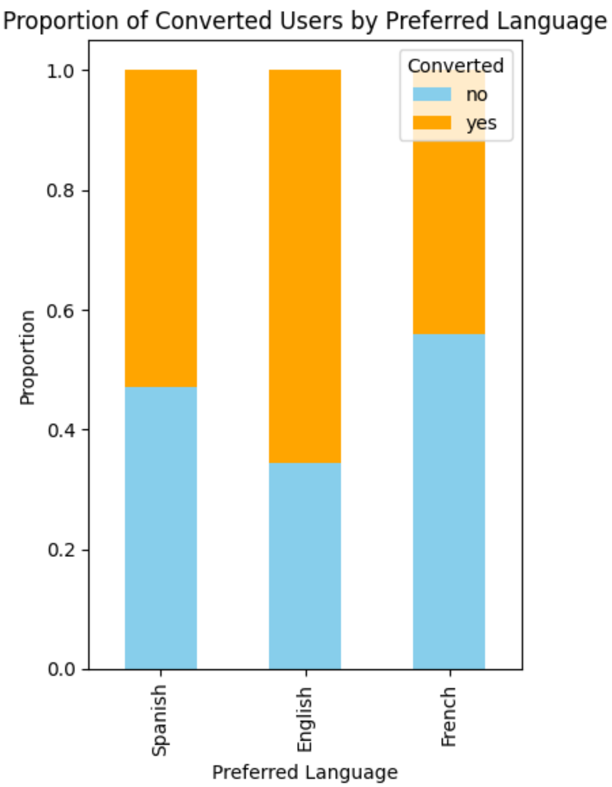
Inference

- Since the p-value (0. 0.00803) < 0.05, there is enough evidence to **reject** the null hypothesis.
- Thus, there is statistically significant evidence that the **new landing page is more effective in converting users into subscribers**.

Link to Appendix slide on details of the test performed (Slide 35)

3. Hypotheses Tested and Results

3. Does the converted status depend on the preferred language?



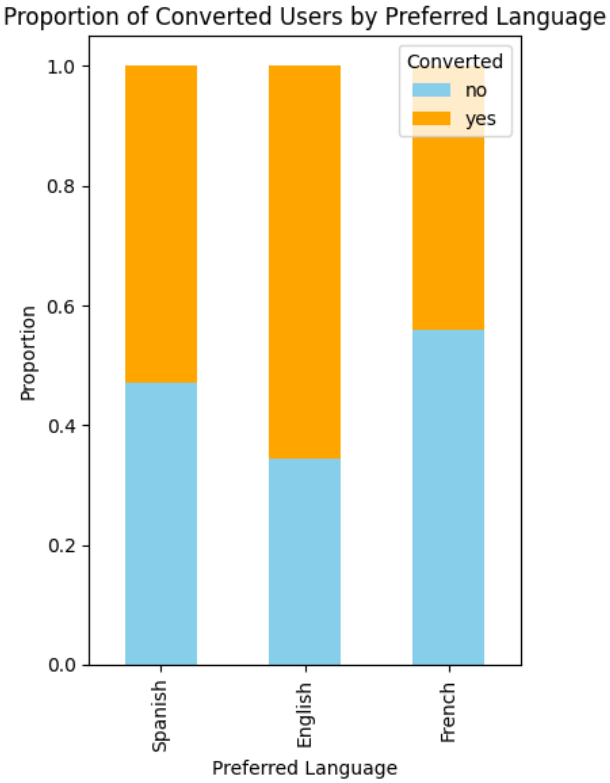
Visual Analysis

- A **stacked bar chart** was used to display the proportion of users who converted (“yes”) and did not convert (“no”) across three language groups: **English**, **French**, and **Spanish**.
- The **conversion rate appears visually higher** for English and Spanish speakers compared to French speakers.
- However, differences among the groups do **not appear strongly pronounced**, indicating a possible lack of significant association.

Link to Appendix slide on details of the test performed (Slide 37)

3. Hypotheses Tested and Results

3. Does the converted status depend on the preferred language?



Hypothesis tested

- It was tested whether **conversion status** (converted vs. not converted) is **associated with users' preferred language** (English, French, Spanish).
- In other words, the test assessed **if language preference has a significant effect** on a user's likelihood to convert (i.e., subscribe).

Test result

- **Chi-Square Test of Independence** was conducted at a **5% significance level** ($\alpha = 0.05$).
- **p-value** = 0.213

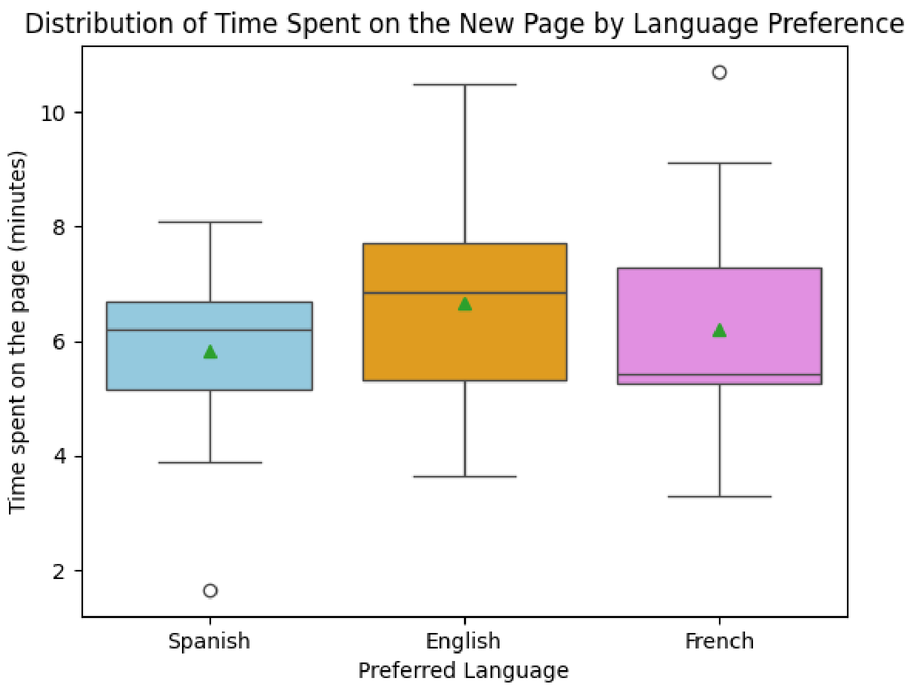
Inference

- Since the **p-value (0.213) > 0.05**, there is **not enough evidence** to reject the null hypothesis.
- There is **no statistically significant evidence** to suggest that conversion status depends on the preferred language.
- Therefore, conversion and language appear to be **independent**.

Link to Appendix slide on details of the test performed (Slide 37)

4. Hypotheses Tested and Results

4. Is the time spent on the new page same for the different language users?



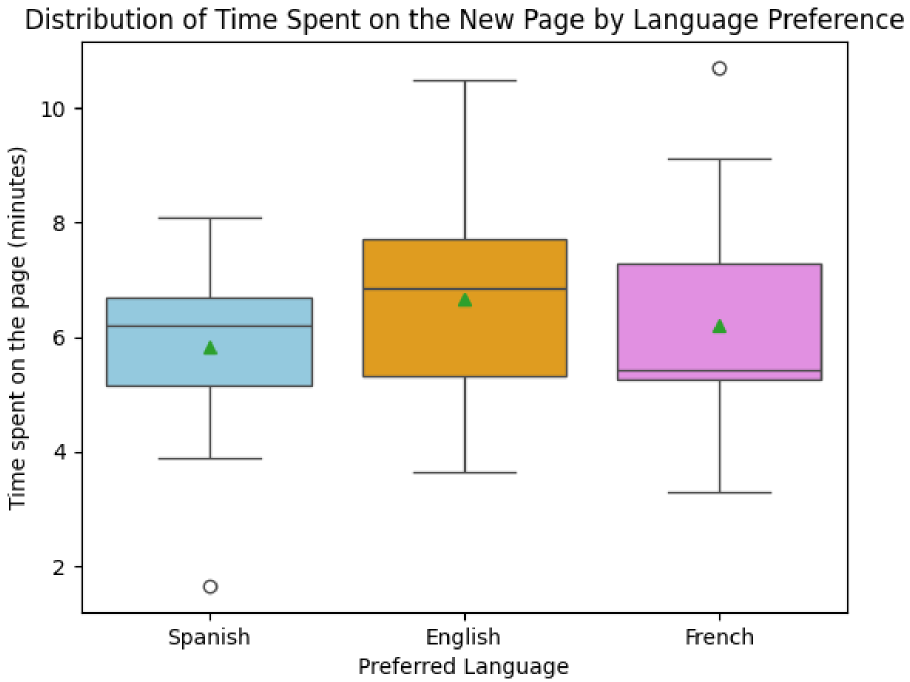
Visual Analysis

- **English users** have the highest median and highest 25th and 75th percentiles, indicating stronger and more consistent engagement.
- **Spanish users** are more consistent (smallest IQR), but their median time spent is lower than English.
- **French users** have the lowest median time spent and the widest variability (largest IQR), with a higher mean than median suggesting right-skewness.
- **Outliers**:
  - **Spanish** users have a **low outlier** (short time spent).
  - **French** users have a **high outlier** (very long time spent).
- Overall, English users spend the most time on the new page, while French users show both lower engagement and greater variability. However, the **means across the three languages show little variability**.

Link to Appendix slide on details of the test performed (Slide 39)

4. Hypotheses Tested and Results

4. Is the time spent on the new page same for the different language users?



Hypothesis tested

- It was tested whether **at least one** language group has a **different mean time spent** on the **new page** compared to the others.

Test result

- **One-Way ANOVA** (Analysis of Variance) test at a **5% significance level** ($\alpha = 0.05$).
- **p-value** = 0. 432

Inference

- Since the **p-value** (0. 432) > 0.05, there is **insufficient evidence** to reject the null hypothesis.
- Thus, the time spent on the new landing page is **statistically similar** across all language groups.

[Link to Appendix slide on details of the test performed (Slide 39)]

APPENDIX

Data Background

E-news Express, an online news portal, is analyzing user engagement to address a decline in new subscriber growth. To improve user experience, a **new landing page** was designed featuring updated layout and content.

A **randomized experiment** (A/B test) was conducted:

- 100 users were split equally between a **control group** (old page) and a **treatment group** (new page).
- User interactions were tracked to measure time spent on the page, subscription conversions, and language preferences.

The **objective** is to statistically determine whether the new landing page improves user engagement and conversion rates compared to the old page, while also analyzing user behavior, understanding user interests, and identifying factors such as language that could further enhance engagement and boost subscription rates.

Data Contents

- **user_id**: Unique identifier for each website visitor.
- **group**: Indicates if the user is in the control (old page) or treatment (new page) group.
- **landing_page**: Identifies whether the landing page served is the old or new version.
- **time_spent_on_the_page**: Time (in minutes) the user spent on the landing page.
- **converted**: Whether the user subscribed to the portal after visiting the page (Yes/No).
- **language_preferred**: Language selected by the user to view the landing page (English, French, Spanish).

Limitations of the Study

- **Small sample size** (n = 100) may limit generalizability about the user population.
- **Unclear sampling method** — possible selection bias (e.g., from a specific region, traffic source, or time zone).
- **Only short-term behavior** was analyzed (no data on long-term engagement or retention).
- **Only language preference** was considered — other user attributes (e.g., age, interests, device type) not included.
- Tested only **one version** of the new landing page — no exploration of alternatives.

1. Hypothesis Testing Details

1. Do the users spend more time on the new landing page than on the old landing page?

Null and alternative hypotheses

- Null Hypothesis: Users **do not spend more time** on the new landing page compared to the old one.

  $H_0 : \mu_{new} \leq \mu_{old}$

- Alternative Hypothesis: Users **spend more time** on the new landing page compared to the old one.

  $H_a: \mu_{new} > \mu_{old}$

Hypothesis Test selected

- Since this is a **one-tailed test** concerning **two population means from two independent populations** and the **population standard deviations are unknown**, the appropriate test is **one-tailed Welch's two-sample t-test** (independent samples t-test without equal variance assumption).

p-value obtained = 0.000139

Inference

- Since the p-value (0.000139) < level of significance (α=0.05), there is **enough evidence** to reject the null hypothesis.
- There is **strong statistical evidence** that users **spend more time** on the new landing page compared to the old landing page.

Link to slide on summary of the test performed (Slide 22)

1. Hypothesis Testing Details

1. Do the users **spend more time** on the **new landing page** than on the **old landing page**?

Computational/ Mathematical Details

Assumptions:

- **Continuous** and approximately **normally** distributed data
- **Independent** and **randomly** selected samples
- **Unequal** population standard deviations (confirmed by F-test)

Summary Statistics:

- **Sample sizes:**

  $n_{old} = n_{new} = 50$

- **Sample means:**

  old page: $\overline{x}_{old} = 4.53$

  new page: $\overline{x}_{old} = 6.22$

- **Sample standard deviations:**

  old page: $s_{old} = 2.58$

  new page: $s_{new} = 1.82$

Checking the Variance with F-test for Equality of Variances:

- Since the population variances are unknown, an **F-test** was conducted to determine whether the standard deviations of the two samples can be assumed equal.

Test results

- F-stat = 2.01
- p-value = 0.0161

Inference

- Since the p-value (0.0161) is **< 0.05**, the **null hypothesis of equal variances is rejected**.
- This indicates that the variances are **unequal**, and thus, **Welch's t-test** (a version of the independent two-sample t-test that does not assume equal variances) is conducted.

[Link to slide on summary of the test performed (Slide 22)]

2. Hypothesis Testing Details

2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?

Null and alternative hypotheses

- Null Hypothesis: The conversion rate for the new page is less than or equal to the conversion rate for the old page.

  $H_0 : p_{\text{new}} \leq p_{\text{old}}$

- Alternative Hypothesis: The conversion rate for the new page is greater than the conversion rate for the old page.

  $H_a: p_{\text{new}} > p_{\text{old}}$

Hypothesis Test selected

- Since this is a **one-tailed test** concerning **two population proportions** from **two independent populations**, the appropriate test is **Two Proportion Z-test**.

p-value obtained = 0.00803

Inference

- Since the p-value (0.00803) < level of significance (α=0.05), there is **enough evidence** to reject the null hypothesis.
- There is **statistical evidence** that the new landing page is more effective in converting users into subscribers than the old landing page.

Link to slide on summary of the test performed (Slide 24)

2. Hypothesis Testing Details

2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?

Computational/ Mathematical Details

Assumptions:

- Dependent variable follows a **binomial** distribution
- Two groups are from **independent** populations
- **Random** sampling was used
- Both np and n(1-p) ≥ 10, so the binomial distribution can be approximated by a **normal** distribution

Summary Statistics:

- **Sample (n_1, n_2):**
  - n_1 = 50 (new page)
  - n_2 = 50 (old page)
- **Conversion proportions ($\hat{p}_1$, $\hat{p}_2$):**
  - $\hat{p}_1$ = 33/50 = 0.66
  - $\hat{p}_2$ = 21/50 = 0.42

[Link to slide on summary of the test performed (Slide 24)]

3. Hypothesis Testing Details

3. Does the converted status depend on the preferred language?

Null and alternative hypotheses

- Null Hypothesis (H_0) : The conversion of users into subscribers **does not depend** on the preferred language.
- Alternative Hypothesis (H_a): The conversion of users into subscribers **depends** on the preferred language.

Hypothesis Test selected

- Since this problem involves **two categorical variables** (Conversion Status and Language Preferred), the appropriate test is **Chi-Square Test of Independence**.

p-value obtained = 0.213

Inference

- Since the p-value (0.213) > level of significance (α=0.05), there is **insufficient evidence** to reject the null hypothesis.
- Thus, the **conversion status appears to be independent** of users' preferred language based on the current dataset.

Link to slide on summary of the test performed (Slide 26)

3. Hypothesis Testing Details

3. Does the converted status depend on the preferred language?

Computational/ Mathematical Details

Assumptions:

- Both variables are **categorical** (conversion status and preferred language).
- The expected frequency in each cell of the contingency table is **at least 5** (to ensure test validity).
- The data was collected through **random sampling** from the population.

Summary Statistics:

- The contingency table summarizes the counts of converted and non-converted users across language groups.

Table 4. Contingency table for conversion status vs language preferred

	Preferred Language		
Converted	English	French	Spanish
No	11	19	16
Yes	21	15	18

- Analyzing the relationship between:

  Conversion Status (Converted = Yes/No)

  Language Preferred (English, French, Spanish)

- **Total sample size**: 100 users
- The **Chi-Square test of Independence** is based on the observed frequencies in this table.

Link to slide on summary of the test performed (Slide 26)

4. Hypothesis Testing Details

4. Is the time spent on the new page same for the different language users?

Null and alternative hypotheses

- Null Hypothesis (H_0): The mean time spent on the new page is the **same** across all language groups.
- Alternative Hypothesis (H_a): **At least one language group** has a **different** mean time spent on the new page.

Hypothesis Test selected

- Since the questions involves **comparing means across more than two groups**, this is a **One-Way ANOVA (Analysis of Variance) test**.

p-value obtained = 0.432

Inference

- Since the p-value (0.432) > level of significance (α=0.05), there is **insufficient evidence** to reject the null hypothesis.
- Thus, the mean time spent on the new page appears to be **statistically similar across English, French, and Spanish users** based on the current data.

Link to slide on summary of the test performed (Slide 28)

4. Hypothesis Testing Details

4. Is the time spent on the new page same for the different language users?

Computational/ Mathematical Details

Assumptions:

- The populations follow a **normal** distribution (confirmed by **Shapiro-Wilk' test**).
- The population **variances are equal** across groups (confirmed by **Levene's test**).
- The samples are **independent** and drawn **randomly**.

Summary Statistics

Table 5. Mean time spent on the new page according to the preferred language.

Language Preferred	Mean Time Spent on New Page (minutes)
English	6.664 minutes
French	6.196 minutes
Spanish	5.835 minutes

[Link to slide on summary of the test performed (Slide 28)]

4. Hypothesis Testing Details

4. Is the time spent on the new page same for the different language users?

Assumption 1: Normality (Shapiro-Wilk Test)

- Null Hypothesis (H_0): The time spent on the new page **follows a normal distribution**.
- Alternative Hypothesis (H_a): The time spent on the new page **does not follow a normal distribution.**

Test Results

- p-value (English) = 0.886
- p-value (French) = 0.321
- p-value (Spanish) = 0.090

Inference

- Since all p-values > 0.05, there is not enough evidence to **reject** the null hypothesis.
- The assumption of **normality is satisfied** for all groups.

Assumption 2: Equality of Variances (Levene's Test)

- Null Hypothesis (H_0): All group variances are **equal**.
- Alternative Hypothesis (H_a): At least one group's variance **differs**.

Test Result

- p-value = 0.467

Inference

- Since the p-value (0.467) > 0.05, there is not enough evidence to **reject** the null hypothesis.
- The assumption of **equal variances is satisfied**.
- Both **normality** and **homogeneity of variances** assumptions are satisfied, supporting the use of One-Way ANOVA.

Link to slide on summary of the test performed (Slide 28)

Happy Learning !